# Shubham Sarkar

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## Summary

Machine Learning Engineer with hands-on experience in building and productionizing AI models and applications. Developed an AI resume-optimization platform serving over 500 users using Python, FastAPI, and PostgreSQL, while implementing MLOps practices for model reliability and performance. Specializes in Natural Language Processing (NLP) and deep learning, with a proven track record of improving model accuracy by 10% through rigorous analysis and evaluation methodologies.

## Experience

**Alignerr** — *Machine Learning Engineer* *Jan 2026 – Apr 2026*
*Remote*

- Evaluated frontier LLM performance across 100+ GitHub repositories by analyzing AI agent code fixes using a 15-metric rubric that included correctness, reasoning, and code quality.
- Identified recurring failure patterns such as hallucinations and incomplete task execution, contributing to a 10% enhancement in model accuracy through root-cause analysis.
- Annotated and analyzed over 100 airline pilot speech data rows, pinpointing aviation-specific terminology and action words to enrich training data quality for speech AI models.

**Berger Paints** — *Engineering Trainee* *June 2026 – July 2026*
*Kolkata, India*

- Gained hands-on exposure to end-to-end paint manufacturing operations, including production planning, raw material handling, process optimization, and quality control.
- Acquired hands-on experience in procurement, inventory management, and warehouse management systems (WMS), including material sourcing, stock control, warehouse operations, and supply-chain processes.
- Analyzed manufacturing and supply chain operations by documenting key operational insights and preparing a comprehensive technical report with process flow analysis and actionable recommendations.

## Projects

**TailorCV** — *Python, FastAPI, LLM, Javascript, PostgreSQL* *January 2026*
Live Demo | GitHub

- Co-developed and scaled an AI resume-optimization platform serving 500+ users across India, owning full-stack development with Python/FastAPI and Neon PostgreSQL across ATS scoring, resume tailoring, portfolio hosting, and AI mock interviews.
- Engineered the core ATS scoring and resume-tailoring engine using a deterministic 100-point audit across 22 weighted checks, with AI rewriting designed to maintain factual accuracy and PDF link-annotation recovery that preserves project, GitHub, and certification URLs across 22 ATS-safe templates.
- Implemented dual payment infrastructure using Razorpay for domestic and Polar for international customers, along with usage quotas, plan gating, and subscription lifecycle management.
- Developed and deployed a Chrome extension for resume tailoring directly from job postings across 15 job boards including LinkedIn, Indeed, Naukri, Greenhouse, Lever, and Workday, reducing per-application tailoring time from 10 minutes to under 60 seconds; automated end-to-end testing with Playwright.
- Set up platform monitoring and growth analytics using Sentry, GA4, and PostHog, with transactional email via SMTP and one-click portfolio publishing through Netlify.

**Youtube Sentiment Analysis** — *TensorFlow, NLP, AWS EC2, Scikit-learn* *October 2025*
Google Drive | GitHub

- Engineered a Flask REST API to deliver real-time sentiment predictions and word-cloud visualizations to a Chrome extension client, managing JSON I/O and CORS for cross-origin requests.
- Trained and evaluated a LightGBM sentiment classifier on 37,800 labeled YouTube comments, achieving 78% test accuracy across three sentiment classes through TF-IDF feature engineering.
- Designed a DVC-orchestrated pipeline (ingestion → preprocessing → training → evaluation) with an MLflow model registry on AWS EC2, ensuring reproducible inference across six tracked experiments.

**Marketing Analytics** — *Python, Excel, SQL, PowerBI, Data Visualization* *Mar 2026 - Apr 2026*
GitHub

- Retrieved and consolidated marketing data from various sources, including CSV files and SQL databases, ensuring data consistency and accuracy.
- Conducted comprehensive data analysis and sentiment analysis using Python to uncover customer opinions, engagement trends, and campaign effectiveness.
- Developed an interactive Power BI dashboard to visualize key performance indicators, facilitating data-driven marketing decisions and actionable business insights.

## Education

**Jadavpur University** — *Bachelor of Technology* *Nov 2023 – Dec 2027*
*CGPA: 7.9*

**Hariyana Vidya Mandir** — *Higher Secondary* *Apr 2020 – Apr 2022*
*Percentage: 90%*

## Skills

**Languages:** Python, SQL, JavaScript, HTML, CSS
**AI/ML:** Machine Learning, Deep Learning, Natural Language Processing (NLP), LLM (GPT, Gemini), Retrieval Augmented Generation (RAG), TensorFlow, PyTorch, scikit-learn, Langchain, MLflow
**Frameworks/Libraries:** FastAPI, Flask, SQLAlchemy, Matplotlib, Pandas, NumPy, Streamlit
**Databases:** MySQL, PostgreSQL, Vector Databases (Chroma DB)
**Tools & Platforms:** Power BI, Excel, Git
**Cloud & DevOps:** Amazon Web Services (EC2, S3, ECS, SES) Docker, Linux
**Other Technical Skills:** REST APIs Data

## Certifications

- Leetcode SQL 50, Leetcode, 2025 | Google Drive
- Internship Training Certificate, Berger Paints Rishra, 2026 | Google Drive

## Leadership & Activities

**Core Member** — *Entrepreneurship Cell, Jadavpur University* *May 2024 – Jan 2026*

- Co-organized national entrepreneurship events E-Summit 2025 and Hult Prize 2025, engaging over 5,000 registrants and 1,000+ participants.

**Core Member** — *FinClub, Jadavpur University* *June 2024 – Jan 2026*

- Organized workshops on investment, financial planning, and market fundamentals for over 100 students, promoting data-driven financial decision-making.